

Parameterizing the Temporally Varying Autocorrelation of Speech

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Abstract

Parameterizing stationary signals based on their autocorrelation is a well-developed and understood technique. Less well understood is how to parameterize signals which are not stationary, even though there is a fair amount of literature devoted to this topic. In this paper, we look at a technique for parameterizing the variation of the autocorrelation of signals: linear interpolation of the autocorrelation coefficients. In order to extend this method to the development of all-pole filters, we use linear interpolation of the principal component of the variance of the autocorrelation coefficients as a means of extending the normal (Yule-Walker) equations to time-varying filters.

Index Terms

Time-varying Autocorrelation, TVAR, time-varying filters, PCA, nonstationary.

I. INTRODUCTION

STATIONARITY is a convenient assumption which allows a class of signals to be treated in a mathematically tractable manner. The class of signals to which it has been applied in the past is far greater than is theoretically justified. This practice has been somewhat amended by the adoption of the assumption of *local stationarity*: namely, a given signal, even if it is not stationary over a long interval, is stationary for a brief ‘window’ of time. We believe that this practice is just a workaround, and that it would be beneficial to develop methods which are more suited to the data under analysis. This paper is an examination of a method of characterizing nonstationary (time-varying) data models and a method of extending this time-varying autocorrelation to the development of nonstationary filters.

A general AR model can be represented in terms of autoregressive (up to the model order, p) as follows:

$$y(t) = \sum_{i=1}^p a_i y(t-i) + \epsilon \quad (1)$$

In this model, the parameter ϵ is generally assumed to be white noise. The obvious extension of this model to the time-varying case is achieved through time-indexing the model parameters, as follows:

$$y(t) = \sum_{i=1}^p a_i(t) y(t-i) + \epsilon \quad (2)$$

Clearly, we now have the rather complicated task of evaluating the coefficients as parameterized by time. Most of the work on this temporally-varying autoregressive (TVAR) problem ([1], [2], [3]) follows the pioneering work of Subba Rao ([4]). In that paper, he modeled the evolution of the ARMA model parameters with a Taylor expansion up to the second order, so that the functions represented by $a_i(t)$ and $b_i(t)$ are constrained to be quadratic functions of time (i.e. $a_i(t) = a_{i1} + a_{i2}t + \frac{1}{2}a_{i3}t^2$). This work was repeated in [1], and expanded to cover several different basis functions (i.e. not only polynomial approximation) in [3].

In this work, we will determine the model parameters using the Instantaneous Estimate of Autocorrelation (IEAC, [5]). This is a time-varying estimate of the autocorrelation of a given signal. It may be seen

as an estimate of the autocorrelation of a signal that is based on a sliding (temporal) window. Implicit in this formulation is the assumption that the autocorrelation changes ‘relatively slowly’ as a function of time, so that it is not greatly biased by the length of the temporal window over which it is averaged. Grenier calls a similar approach “The simplest assumption which can be made” ([3]), and correctly asserts that the limitation of this method is that if we want to get detailed spectral resolution, we must use a long window to average the autocorrelation coefficients, which in turn will misrepresent signals that have a quickly changing autocorrelation function. While this criticism is justified, we feel that this tradeoff is just a restatement of Heisenburg’s well-known Uncertainty Principle, about which very little can be done for this or any other method of estimation.

We base our analysis of the time-varying autocorrelation of a signal using the IEAC, which we transform into the autoregressive model coefficients using the Yule-Walker equations. Given a signal $x(n)$ of length N , we define its time lagged vector ($\mathbf{x}_l$) and IEAC vector ($\mathbf{x}_c(n)$) as follows:

$$\mathbf{x}_l(n) \triangleq [x(n) \ x(n-1) \ \dots \ x(n-M)] \quad (3)$$

$$\mathbf{x}_c(n) \triangleq [x(n)^2 \ x(n)x(n-1) \ \dots \ x(n)x(n-M)] = x(n)\mathbf{x}_l(n) \quad (4)$$

Where M denotes the number of lagged values of the vector. We assume that $x(n) = 0$ for $n < 0$, thus we form a vector representing the signal as $\mathbf{x}_c$ which is of dimension $N \times (M+1)$. In order to reduce the variance of this signal, it may be desired to smooth it with a window such as the normalized Hanning. A more thorough analysis of the IEAC is given in [5], and will not be repeated here.

As the estimation of the moving average model parameters (b) is a difficult non-linear task, we will only concern ourselves with the autoregressive (a) parameters; thus, we will be modelling a signal with an all-pole filter, whose solution can be had (without matrix inversion) through the application of the Levinson-Durbin recursion to the normal equations ($\mathbf{R}_x \mathbf{a} = \mathbf{d}$):

$$\begin{bmatrix} r_x(0) & r_x(1) & \dots & r_x(P) \\ r_x(1) & r_x(0) & \dots & r_x(P-1) \\ \vdots & \vdots & \ddots & \vdots \\ r_x(P) & r_x(P-1) & \dots & r_x(0) \end{bmatrix} \begin{bmatrix} 1 \\ a_1 \\ \vdots \\ a_P \end{bmatrix} = \begin{bmatrix} |d_0|^2 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad (5)$$

Another way of estimating the all-pole predictor coefficients $\mathbf{c}$ is to directly estimate them using a least-squares solution to the following:

$$\begin{bmatrix} x(t) \\ x(t-1) \\ \vdots \\ x(p) \end{bmatrix} = \begin{bmatrix} x(t-1) & x(t-2) & \dots & x(t-p) \\ x(t-2) & x(t-3) & \dots & x(t-1-p) \\ \vdots & \vdots & \ddots & \vdots \\ x(p-1) & x(p-2) & \dots & x(0) \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_p \end{bmatrix} \quad (6)$$

Since our autocorrelation coefficients are assumed to vary with time, we model them here using linear regression. This leads to the following (modified) form of the Yule-Walker equations:

$$\begin{aligned} [\mathbf{R}_1 + \mathbf{R}_2 t] \mathbf{a} &= \mathbf{d} \\ \mathbf{a} &= [\mathbf{R}_1 + \mathbf{R}_2 t]^{-1} \mathbf{d} \end{aligned} \quad (7)$$

Computing the matrix inverse present in equation 7 at every time step is clearly untenable. In order to proceed, we formulate the autocorrelation matrices in terms of their eigendecomposition:

$$\mathbf{a} = [\mathbf{Q}\mathbf{\Lambda}_1\mathbf{Q}^H + \mathbf{Q}\mathbf{\Lambda}_2\mathbf{Q}^H t]^{-1} \mathbf{d} \quad (8)$$

$$\begin{aligned} &= [\mathbf{Q}(\mathbf{\Lambda}_1 + \mathbf{\Lambda}_2 t)\mathbf{Q}^H]^{-1} \mathbf{d} \\ &= \mathbf{Q}(\mathbf{\Lambda}_1 + \mathbf{\Lambda}_2 t)^{-1} \mathbf{Q}^H \mathbf{d} \end{aligned} \quad (9)$$

In order for these equations to hold, we must have an identical eigenvalue matrix $\mathbf{Q}$ in equation 8 for both $\mathbf{R}_1$ and $\mathbf{R}_2$ in equation 7. This is not, in general, possible; to proceed, we will determine a common

eigenvalue matrix (which we will denote $\tilde{\mathbf{Q}}$), and then determine eigenvalue matrices $\mathbf{\Lambda}_1, \mathbf{\Lambda}_2$ which most closely enables approximation of the autocorrelation matrices $\mathbf{R}_1$ and $\mathbf{R}_2$. As long as we can develop a suitable $\mathbf{\Lambda}$, we can easily describe the trajectory of the all-pole filter coefficients (since taking the matrix inverse of a diagonal matrix is relatively trivial).

In order to model the evolution of the eigenvalues of the system (so that we approximate the evolution of the autocorrelation matrices), we must pick a matrix $\tilde{\mathbf{Q}}$ which is able to (approximately) decompose the estimated autocorrelation matrices at every instant of the signal. We choose the matrix which is formed by the eigendecomposition of the correlation matrix of the IEAC signal. As such, the column vectors of $\tilde{\mathbf{Q}}$ represent the directions of maximum variance of the autocorrelation of the signal, $\mathbf{x}$. It is expected that this choice of $\tilde{\mathbf{Q}}$ will allow us the best approximation of the evolution of the autocorrelation matrix (with respect to time), since it potentially allows us to best parameterize the change (variance) in autocorrelation.

According to these observations, and recalling equation 4, we obtain $\tilde{\mathbf{Q}}$ from the following relation:

$$\mathbf{x}_c^H \mathbf{x}_c = \mathbf{R}_{\mathbf{x}_c} = \tilde{\mathbf{Q}} \mathbf{\Lambda} \tilde{\mathbf{Q}}^H \quad (10)$$

In order to determine eigenvalue matrices $\mathbf{\Lambda}_1, \mathbf{\Lambda}_2$, we utilize the notion of a ‘whitened’ vector $\mathbf{w}_c$ which corresponds to the variation of the lagged vector $\mathbf{x}_1$ along the principal components of $\tilde{\mathbf{Q}}$:

$$\mathbf{w}_1 \triangleq \mathbf{x}_1 \tilde{\mathbf{Q}} \quad (11)$$

$$\mathbf{\Lambda} = E(\mathbf{w}_1^H \mathbf{w}_1) \quad (12)$$

From the equations above, it is trivial to determine a constant estimate of the eigenvalues:

$$\hat{\lambda}_j = \frac{1}{N} \sum_{i=0}^{N-1} w_i(i, j)^2 \quad (13)$$

From this, we form linear regression matrices which parameterize the temporal evolution of the eigenvalues as follows:

$$\begin{bmatrix} w_i(0)^2 \\ w_i(1)^2 \\ \vdots \\ w_i(N)^2 \end{bmatrix} = \begin{bmatrix} 1 & t(0) \\ 1 & t(1) \\ \vdots & \vdots \\ 1 & t(N) \end{bmatrix} \begin{bmatrix} \omega_0 \\ \omega_1 \end{bmatrix} + \begin{bmatrix} \epsilon_0 \\ \epsilon_1 \\ \vdots \\ \epsilon_N \end{bmatrix} \quad (14)$$

Where the desired parameters ω_1 and ω_2 represent the eigenvalues λ_1, λ_2 , and the error term is represented as ϵ_i .

Having obtained these parameters, we return to the problem of determining the model (predictor) parameters $\mathbf{a}$. By using the first two terms of the Maclaurin expansion of c (from equation 5) combined with equation 9, we have that:

$$\begin{aligned} c(t) &= c(0) + \frac{c'(0)}{1!}t \\ &= \mathbf{Q} [\text{diag}(\lambda_1 + \lambda_2 t)^{-1}] \mathbf{Q}^H \mathbf{d} \Big|_{t=0} + \mathbf{Q} \left[\frac{d}{dt} \text{diag}(\lambda_1 + \lambda_2 t)^{-1} \right] \mathbf{Q}^H \mathbf{d} \Big|_{t=0} t \\ &= \mathbf{Q} [\text{diag}(\lambda_1)^{-1}] \mathbf{Q}^H \mathbf{d} + \mathbf{Q} [\text{diag}(-\lambda_2(\lambda_1 + \lambda_2 t)^{-2})] \mathbf{Q}^H \mathbf{d} \Big|_{t=0} t \\ &= \mathbf{Q} \left[\text{diag} \left(\frac{1}{\lambda_1} \right) \right] \mathbf{Q}^H \mathbf{d} + \mathbf{Q} \left[\text{diag} \left(\frac{-\lambda_2}{\lambda_1^2} \right) \right] \mathbf{Q}^H \mathbf{d} t \end{aligned} \quad (15)$$

From this equation, we see that we have a formula for the coefficients of the all-pole filter in terms of time, which is the result of parameterizing the IEAC signal with a linear function. By combining the expressions in equation 15, the linear eigenvalue trajectory as a function of time may be written as:

$$\lambda = \frac{\lambda_1 - \lambda_2 t}{\lambda_1^2} \quad (16)$$

From this, we observe that our method will run into problems when the eigenvalues $\lambda_1 \dots$ In terms of stability analysis, ...

II. METHODS

In order to assess the algorithm proposed in this paper, we analyse several frames of digitized speech. The transition from one speech frame to another is not addressed: our objective here is only to determine the mean square error of the technique presented in this paper with a stationary frame blocking technique (i.e. one which only uses an estimate of the autocorrelation assuming that the signal is stationary).

The speech segment is a recording of the spoken phrase “Noun phrases lack inherent existence”. We use frame sizes of 10, 20, and 30 milliseconds.

III. RESULTS

IV. DISCUSSION

Clearly, other methods of parameterizing the IEAC signal are available, such as using higher order polynomials or other basis functions (such as those described in [3]).

V. CONCLUSION

APPENDIX I

ERRATA

Here we have some more stuff:

$$\mathbf{R} = E(\mathbf{x}^H \mathbf{x}) \quad (17)$$

$$\mathbf{R} \mathbf{q}_i = \lambda_i \mathbf{q}_i \quad (18)$$

$$\mathbf{\Lambda} = \text{diag}[\lambda_1, \lambda_2, \dots, \lambda_M] \mathbf{w}^H = \mathbf{Q}^H \mathbf{x}^H \quad (19)$$

$$E(\mathbf{w}^H \mathbf{w}) = \mathbf{\Lambda} = E(\mathbf{Q}^H \mathbf{x}^H \mathbf{x} \mathbf{Q}) \quad (20)$$

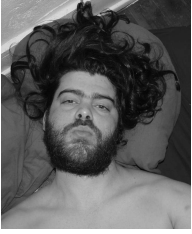
In order to derive diagonal matrices $\tilde{\Lambda}$ which best represent our autocorrelation matrices in terms of our chosen $\tilde{\mathbf{Q}}$ (i.e. $\mathbf{R} = \tilde{\mathbf{Q}} \tilde{\Lambda} \tilde{\mathbf{Q}}^H$), we choose $\tilde{\Lambda}$ to be equal to the diagonal elements of $\mathbf{\Lambda}$, where:

$$\mathbf{\Lambda} = \tilde{\mathbf{Q}}^H \mathbf{R}_x \tilde{\mathbf{Q}} \quad (21)$$

We could also choose $\tilde{\mathbf{Q}}$ to come from the average (stationary) autocorrelation matrix. This approach is not pursued in this paper. In equation 8, we obtain the (unitary) Q matrices by eigendecomposition of the correlation matrix of the IEAC ($\mathbf{C}_{\mathbf{x}_c}$, [5]). Λ_1 and Λ_2 are estimated by projection of the signal onto the Q (principal component) matrix.

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